

ActInf Livestream #019.1 ~ Deeply Felt Affect: The Emergence of Valence in Deep Active Inference

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[Music] hello everyone welcome to the active inference lab and to the active inference live stream this is active inference live stream 19.1 on april 6 2021 welcome to the active inference lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at our links and contact information here this is recorded in an archived live stream so please provide us feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll be following hopefully video etiquette for live streams such as muting when there's noise in our background and raising our hands so we can hear from everyone who wants to speak you can go to this shortened link to see the upcoming streams of different kinds today we're in 19.1 on april 6th and we'll be having a follow-up group discussion next week on april 13th hello steven we're already live so today in active inference 19.1 the goal is really just to learn and discuss about this really awesome paper deeply felt affect the emergence of valence in deep active inference by casper hesp ryan smith thomas parr mika allen carl fristen and maxwell ramstead and we're really appreciative that casper has joined us today and any one of these other authors are free to join us next week or the following in 19.1 we're gonna go over the various parts of the paper in any order that people want to raise them and we can just start with some short introductions and warm-ups that will be culminating in casper and giving a little maybe just introduction or a background and um that will be great to start with so i'm daniel i'm a postdoc in california and i will pass it to sarah i'm sarah i'm a person stuck in berlin because i can't update my visa uh next hello i'm stephen i'm in toronto and uh i'm very much in looking forward to learning more about effect and it's a relationship to active inference and i'm going to pass it to day okay dave douglas i'm in the mountains of the philippines retired from information technology including natural language processing and machine translation background in process philosophy whitehead especially and cybernetics and i will pass to dean if he hasn't spoken yet i'm dean i'm from calgary um retired but really interested in this stuff i don't know who i can pass it to

other than maybe casper i'm not sure sure yeah my name is uh casper i'm a phd student at the university of amsterdam i'm currently doing yeah um kind of being co-supervised by two professors at the university of amsterdam and um one being at the university of uh college london so that's uh the one you probably all know called kristen um it's like my um external uh supervisor and this paper was developed under his um yeah in his lab basically with a bunch of really wonderful people actually my co-first author is not here today but i am aware that he has been brian smith has been with you on many of these sessions and yeah so i think i'll just go into the intro of the paper and i guess it would be interesting because you can read in the paper itself in a kind of formal motivation i will tell you a little bit more about the story behind how this how this came about when i first entered the group there was most of the modeling that was done was with single layers and there was already some preliminary work on deep temporal modeling and that's essentially working out the implications of this kind of hierarchical models that start to look more like um the one that might have come across some deep learning research but now with the bayesian beijing flavor and one of the things that i was missing personally was a meta cognitive aspect that moves beyond just estimating uh confidence it's one one aspect but then reflecting on that confidence estimate to give you an idea of something that can regulate other parts of the system so we ended up going back and forth on this topic with uh with carl and thomas farr and was a huge help there as well and he's also on the paper essentially thinking about these action models in active implants and i'm not presupposing certain degree of familiarity but i can suppose i can quickly read that essentially the way i've presented it in paper is really meant to be something that you start from zero knowledge about active influence and i kind of build it up in steps so that people are not super familiar with it yeah i guess it's like yeah these these steps essentially the the point is that these diagrams have a very mathematical meaning um a very specific mathematical and when you just present people the whole diagram at once what we call directed acyclical graphs it usually is pretty overwhelming so for this paper i decided to kind of go back and just give the viewer the reader a very stepwise kind of incremental primer we call it and we start with um these first two states that are being linked is the the types of states is hidden states and sensory states and that's really the kind of core and what's interesting is that it's very modular so if you go to the next slide yeah so don't if you just ignore the maps for now but the nice idea here is that these graphs are entirely or it's kind of like a lego box essentially so once you understand the first um piece of the puzzle in m1 where it's inferring hidden states based on sensory states you can basically understand all the other graphs as well to a certain degree i mean at some

point we introduce um links between continuous and discrete state spaces that's where it gets a little bit more complicated mathematically the one thing you can keep in mind when reading these figures is that down stream on the arrows is always about prediction so here for example the prior overridden on the top d predicts um the hidden state below that s so downstream on the arrows we're talking about prediction upstream we're talking about inference so in this case you have you start from the prior make prediction about what the hidden states are a priori and then moving from these hidden states make prediction about what the sensory states are and that's where the likelihood nothing comes in and then so that's one crucial aspect so downstream prediction upstream inference then there's another crucial part that you need to know in order to interpret these figures is that um the circles are variables and the squares are parameters so in this case hidden states and sensory states those are listed in the circles the s and the o those can vary over the course of the simulation and if you set it up in this very simplistic way then the squares are fixed so the values in the prior and the likelihood method are fixed in this very simplistic section once you understand this you can start hooking these things up to each other so we can once we link another circle to a square then this uh the mappings become variable as well and you will see that coming back later but it would now generalize this reading of the figure to the second and so m_2 here exactly the same maps is applied to this way the states evolve over time and so generative model anticipation is using current states to make predictions about future states that's where we come to the third part of these kinds of graphs that's pretty important to know is that to right it's supposed to be the temporal dimension so s_1 is used to make predictions about s_2 and base if you have knowledge about s_2 you can make inferences about what s_1 was the most likely state in the past as well so it works both ways so that's perception and anticipation and obviously entirely reduced to the most sim yes simplistic kind of form that you couldn't come up with essentially something often these models are criticized for being too simplistic well i think that's kind of the beauty in the sense that we obviously start with these components the lego blocks so to speak the individual blocks are pretty simple but once you start hooking them up to each other you get emergent dynamics that are pretty uh hard to already create yeah pretty quickly become hard to wrap your mind around so that's why we're presenting them in a stepwise question i would say if you're really interested in the maps then um you would probably want to go into one somewhat the tutorials for example that ryan smith recently made i can actually send a link to that right now because he works through the actual equations in this step-by-step session cool you can put any links in the youtube live chat so that everybody can see it

otherwise only we'll be able to see it yeah cool thanks for all these clarifying points on figure one and i agree it'd be awesome to have um ryan uh these have been on a few times so yeah um okay question related figure one stephen yeah just one question when you mentioned the predictions and inferences do you tend to think about like the downward predictions across all the components and then the inference is more although it's happening at all components more in terms of like the big block jumps between like what m_1 and m_2 is inferring in relation to each other so you tend to think about it more that way so you look at each piece as a as more of smaller ideas predicting down and linking and then the thought of in these you know broader sense of something that you could get your head around as a human so to speak um it depends on which level of description you're currently focusing but um in the sense in a sense the answer is yes that um once you kind of stick another layer on top of it so to speak that layer can start to um it's basically perceiving uh in a weird way you could see say it's trying to perceive the perceptions and then that allows the system to report on these perceptions so that's like a kind of meta and meta-level representation that gives the system this capacity to reflect on its own on its own inferences on the lower level we have also worked on paper where we apply this to the precision on the likelihood and should probably not get into that too much for now but it's if anybody's interested so let me just i found i've gotten access to the live stream now let me just post say hello and then i'll make you a mod so that you can post links but yeah thank you for adding links to the live stream anyone else have a question on figure one otherwise it'd be nice to i think continue this incremental unrolling of the model and then we'll definitely get to a lot of different topics i think in a broader discussion too i'm in the live chat right now so here's the paper by ryan smith and then there's now you're a moderator so now nice okay steven question on figure one or broader question uh yeah just building on what was just mentioned a second ago with figure one is you know thanks for that clarification and um so like you say this this idea of like a meta level of inference is useful and i think and this came up in the previous um live stream with ryan smith that there seems to be something about the updating between so as well as like going from one chunk to the next chunk so to speak of $m_1 m_2$ and giving that as a way to get a handle on an inference there's like something that can be done with whatever gets message passed up and so from what my understanding that looks like the two ways into the inferential piece so great yeah so that's where um essentially one of the core innovations that we present in this paper is something we like to call deep parametric active um where you're making inferences about the parameters of lower level models and then that is guiding your your expectations and we're going to get into that

more we once we go down this path and let me see so i've managed to get access i think so what does the moderation help me oh you could post links i think people who are moderators cannot post links but we got your links so the tutorial paper and the meta awareness paper are in the live chat so thank you yeah good okay cool um yeah so i just to illustrate the point i think it makes sense to go to go further um because essentially what uh i think it was dean was mentioning or was it dave i couldn't see and it's my own site but um um anyway so and what was mentioned is that yeah you have these lego blocks and you start building your hierarchy so to speak and one of the things that was had already happened by the time i came around was this step presented here where again the circle here with the π inside is a variable and the simple act of connecting this variable to the state transitions on the lower level so basically the variable π which stands for policy starts to dynamically control your beliefs about how to stay how the states of the world equal all the time yes so this is um then controlled by the prior um and this is a really a little bit of a funky aspect of active inference in that formulation because the prior on this on these policies is informed by a the expected free energy the expected free energy i call is funky because it recapitulates the entire generative model down below the expected free energy includes the expectations of the organism about the world so there's this thing that carl and thomas bar had called and the generalized free energy where they reformulate this and impact what's inside the policy variable and in terms of a whole kind of a generative model that mirrors um the perception aspect um don't get need to get too deep into that that's essentially unpacking what uh the expected free energy here does in the model um it's very important because the expected free energy is what introduces the what i call the phenotypic or phenotype congruent action model it's also going to be answering one of the questions that came up i watched the previous live stream that you had on this paper and we'll get to that in a second but we can use it to answer one of the questions that came up about um whether the effective charge uh why it depends only arises when there is a difference between the prior and posterior objection we'll get to it so the expect free energy you can think of it as recapitulating the entire action model with one big difference namely that it's biased by phenotypic outcome practices so it's essentially the phenotypic outcome preferences basically bring into an active inference what the reward function does in reinforcement it does this because it's essentially biasing the way the agent behaves towards the world and trying to match make naked match its preferences that are specified here purely in probabilistic terms so the phenotypic preferences can be something that can be engrained that can be learned it's not really it's pretty agnostic where it comes from um that's because there it has this hierarchical

nature so i have models in which these preferences themselves are state dependent so let's say you're hungry um when you're hungry your preference uh factor for um for food is then is up regulated essentially it makes um more uh a stronger driver of your actions essentially when you're tired then you have a stronger preference against exerting effort so there's these kinds of um these preferences themselves can be made state dependent if you just add another lego block on top and you start having inferences about how hungry you are how tired you are and you can have an organism that's managing its energy resources then there's this last component that we discussed is the e here the e matrix you can think of it as a kind of habitual prior or what they call in mobile in a and they call it model 3 it's just counting the number of times a particular action occurred it's not evaluating it by any any like by any rewards or something like that it's just um i mean i would i actually don't like the term model 3 because in essence it is still model it's just a model that has a separate parameter for every possible action right so just counting it's kind of like when you took your drivers lessons and i remember repeating mistakes not because i liked to make those mistakes but just because i made them before my body and brain kind of stored that information and i ended up repeating those mistakes just because this pathway has been explored before okay so that's like a little bit of unpacking here and the way you can read this diagram is again from left to right so you start with the action model and the expected free energy guiding the a priori expectations and this is a kind of crooked scientist i would say it is biased towards not what you believe the world to be exactly but more towards what you would like the world to be like and or what's if you are currently doing well what you would expect the world to be so it's pretty um it's it's an intentional but yeah like the bias is introduced in that site and then if you look on the other side you get the perceptual evidence it's the free energy you can think of that as a kind of reality check so i can have lots of expectations about what i want the world to be like and where i want to end up or what i want my shoes to be like when i buy them i remember this example from last time um but then i need perceptual evidence to actually see if the shoes actually fit after i bought them let's say um so this perceptual evidence gives a very it's a very crucial component of how how the whole the kind of perceptual feedback without that your your agent would be living in a fantasy world right so we can if there's unless there are any questions we can move to the next slide anyone uh can raise their hand otherwise stephen question on this figure or a broader question okay go for the question this figure just want to ask you one question there with your action model because with expected free energy you often have the risk ambiguity and pragmatic gain as three terms on one side so is it like that the pragmatic gain is built into the

action model implicitly so it's kind of you you've rearranged it slightly with that in mind so that all action has some sort of pragmatic game built into the phenotypical morphology of the animal or something like that yeah so here in the pragmatic part so that's the kind of the see the outcome preferences like your preferred observations essentially is built into its part of the phenotypic risk so if you look at the equation on the left and so the right hand side to the left term for phenotypic risk you see there you take the the difference between the logarithm of your expected outcomes minus your preferred outcomes then in in terms of units of nets so information to your experience so what you're essentially the risk here is just quantify in terms of the the difference between your preferred outcomes and the expected outcomes given a particular action that makes sense cool and it's also interesting it's the natural log of o_{π} so it's conditioned on a policy so it's not like some sort of all by all chess boards all strategies all situations it's very constrained with respect to what actually matters which are in the end the affordances in the niche of the actual agents in the in its scenario so it's really cool to see how that gets baked into the not like a secondary pruning or a heuristic yeah so this this just falls out of calculating and the expected to energy under a certain μ policy and so this g is conditions on policies in that sense μ and one interesting aspect here is that if you make the generative model more sophisticated then this g is going to become more sophisticated as well so you can add if you add learning or parameters of let's say the likelihood then the aspect of your also expands accordingly and it ends up having a term for active learning so your agent can anticipate basically that exploring the environment can allow u to reduce uncertainty about these perceptual μ mappings so then you get an agent that's intrinsically curious about the world you can do the same with the b matrices or the transition matrices so you get an agent like a child who kind of just experiments exploring different states of the world and then starts picking up on different kinds of control that it has and can even at some point can even learn to anticipate that if it's μ experiments in a certain way it might be able to learn new things about these methods so this is this is currently in not in this particular model because if you remember to do that you need to add variables to the particular methods and the nice thing about this lego box of active inference that speaks that you're completely free to just to add μ to make anything variable as long as you can yeah as long as you specify certain the probability distributions you can do the same trick again and again in different on the street in continuous state spaces so i think this is a nice segue into oh here you first introduced in table three some of the equations μ here we have table three up on the screen or do you want figure three no i think we can go on to i mean unless somebody wants to go deeper into this

but i i think you already kind of discussed this yep it was really helpful to see the table anyway laid out this is work yeah this is actually not the way i delivered the table to the journal it's just that they have three i mean it's crazy you pay them this money to fix it you know but then they make it worse so like basically they had like a new what layout and their webplay items and messes up the most of the figures and so when you want to read it i would really recommend downloading the pdf and not using their web oh yes this is their web layout it it definitely makes sort of a false equivalence or a way of making it look like there's something missing on one side or the yeah so just read the pdf and it's unfortunate because it's supposed to be their job well here we are on figure three and so we have just the figure three and then we also have that part with the ac if you want to kind of highlight that because i know that's something we really want to understand yes so um just we're going to continue our our um building this um different we're adding these different blocks together and now what we add on top of it is we have g this is the recapitulation of the entire action mode we add a variable on top of it that modulates the extent to which i rely on this and so the precision gamma can take any barrier value between 0 and infinity and and it just tells me when i predict what's going to happen in the world and i anticipate what's going to happen um how much can i rely on my action model that's biased by my preferences so this is not the same as um i mean yeah that's kind of where it comes into into um story why and it's this effective charge that we're talking about ends up being depending on the difference between pi and prior and the posterior so essentially what this precision is tracking is tracking the match between the predicted value you can say of g or predict for the actual policies that i end up selecting um the actual state transitions of the worlds and how well they are predicted by my action and this is my crooked scientist model right so it's like the max action model that's trying to realize preferences and effective charge is essentially the kind of um we call it like that because later on it's getting a special role and when you go to hierarchical setup we add one more lego block on top of this but essentially um whenever my um whenever um so if you unpack this effective charge equation you see that it's calculating the difference between my prior policy vector that's just based on my preferences my action model um and the posterior and that's the one that happened that i have after i integrate perceptual evidence so and then that difference so that's where it's a little bit hard to imagine maybe but basically um the difference between these two is kind of like the rate of change you could say over time of my beliefs and then you take the dot product of that with the expected free energy and what what you essentially want to know is um after integrating perceptual evidence did they get closer to matching the expected free energy the

distribution anticipated by that or did i get further away so that's why i call this phenotypic progress if i get if the expected pure energy is a bad predictor then this term ends up being uh negative if it's a good predictor it ends up being positive and i mean there's like a little bit of a play with signs there and we just define it in a way that it's positive the positive value is a good thing that's like always a little bit confusing because in some different fields they use there's even fields where the free energy is defined in the opposite way so the whole story clips on his head that's why you always have to be careful how you how you define it locally in each paper and yeah when you come from a different field you have to you know make sure that you've got your signs in your signs in a row so to speak um yeah so if i can say that one major field there difference is that physicists often go downhill gradient descent optimization we're going to minimize the loss function and biologists often talk or think about hill climbing and fitness peaks and optimization maximization and congruence in the best possible world that kind of pangolos world so totally agreed and especially when there's a lot of natural logs and negative signs and differences and divergences that can only be positive and things that are strictly negative or bounded but when it's something where it really needs to be walked through slowly because it's so easy to get tripped up in how these things flip back and forth so yeah really important notes to kind of just go slow and really make sure that we're using the right tools to establish that we're connecting the right qualitative ideas with these variables because it's not just like you throw up the equation and then it's going to be easy to define what each of the variables are for real systems yeah and on that note we basically yeah we have um interpreted this comma as uh in the paper as a type of subjective fitness and that's because literally is tracking the degree to which your um preference biased action model is fitting with the actual perceptual evidence that you're getting back and and that's why the effective charge can only be non-zero if there's some mismatch between your prior and your posterior beliefs about policies because um there's only when there there's some mismatch between what you expected when you got and it makes sense to talk about updating your beliefs about procedure and can be in a positive direction or in a negative direction so my fit can be actually better than expected and or worse than expected that's why why describe in the in the abstracts that this lens is assigned to these predictions essentially because if you think about just in terms of predictions it's kind of well prediction is either wrong or very wrong okay so then i mean if you want to know um if you go to this meta level where you're estimating how wrong am i so is my prediction getting better or worse that's that's where the sign comes into into the story and where you can talk about um improvements or uh

improving or worsening state of the world speed and state of yourself and this really applies to any kinds of states by the way so it can um it can be about internal states tracking and as i said hunger and fatigue but it can also be about the objective states of other specifics so um this kind of subjective fitness can be linked to any arbitrary state that of interest and if you go to the next level then the state of inference can be an observation for the next level so that's hierarchical trick and is i think the power of it is that yeah we have freedom to do what we want to do with our lego blocks depending on this system of interest but we try to keep it simple while we're still just demonstrating possibilities but then the implications get really really big once you start to release the simplicity constraints and you just go wild on constructing models hierarchical models with different you can have expect your energies at different levels you can have these kind of precision terms at different levels so you can have competing um affected charges you can have the effective charge from your lower level let's say you're trying to fight your addiction or something like that the lower level parts of your system are like kind of great and generating this craving and the higher level parts are like trying to generalize and generate this kind of um self-actualization or something like that where you basically end up having conflicting drives and so i mean that's where where we can go but to get there we need to move to add states on top of this and i think and that's there yeah if there are questions yep i'm gonna ask a question from the live chat and then anyone else who has didn't spoke yet or asked a question and then we'll ask a question about um anything else we've talked about steven so um in the chat someone asks can we relate affective charge with motivation as motivation can also affect policy so where do we how do we think about motivation in this kind of a model is it an affective charge is it effective charge yeah so um motivation in that sense is a very i mean kind of fuzzy concept right but um if you think about the gamma term here it's modulating the extent to which you're relying on your action model and the c the preference factor kind of modulating the um the extent to which you're driven by by your preferences and combining them together essentially means that um you're if you have um let's say a high and yeah you need high confidence in your action model to be actually allow it to influence your your expectations so in that sense there is definitely a link with motivation but then again you also need preferences that motivates you to move towards them right so what's kind of interesting is if you take away the preferences then you can have an agent that's motivated entirely by epistemic family so i suppose to some extent um it's kind of interesting because you get precision starts to modulates curiosity and so that's a different kind of motivation anyway i can say much more because it's such a fuzzy concept but very

true i reward motivation curiosity motivation there's expressivity in the model to actually talk about motivation with respect to specific framings of it yeah and and um the way i've recently thought about it is also that if you add a higher level and you connect it to this preference matrix um you can basically modulate the extent to which you're motivated by different kinds of outcomes and that means that um you're if you're like a monitoring on a higher level to which extent you're currently satisfied for example um so you're you're hungry so you upregulate your preference for eating food and then you get your food and you get feedback intercepted feedback that you're you're you're satiated and then you're open for um more for epistemic drives let's say so um there's an extended yeah a sense in the various types of motivation have to be balanced by a higher level this current figure that is not able to do that in a if i mean our whole lego box can definitely do that but the current figure i think is still um doesn't have dynamic preferences i think that's what you need to get a more satisfying account for motivation in general thanks for that awesome response all right stephen and then anyone else who raises their hand just one question you mentioned dot product with expected free energy and just just if you could just clarify that a little bit because i think you've got something to do with the precision you've got the the larger free energy piece so if you just clarify that would be helpful um yeah so this effective charge term is not something we came up with right it's something that comes out of the maps when you start with um this uh gamma distribution um sorry the capital gamma distribution is um particular mathematical um yeah shape and exponential natural exponent based on natural exponent and when you use that as a probability distribution so it has this kind of tail that runs goes towards infinity um the expectation value of this tail is regulated by what's called here the temperature parameter and also called the rate parameter in other contexts but essentially the temperature parameter kind of regulates um how um how strongly this yeah the distribution is shifted towards zero and essentially what's what happens is that this when you use that as the probability distribution and you try to optimize the best this better parameter to best fit what's happening your you can do this for energy minimization to do that and what you get out of it kind of for free and once you write that down in probabilistic terms you get this objective charge there and what i was talking about is essentially an interpretation of what okay this is what we got out when we took the generative model minimized let it minimize variational free energy on these this battery parameter then we just it just kind of spits out this effective charge too and i was yeah then it's a matter of interpreting like what does this term mean in mathematical sense and essentially what it's doing is taking dot product between two vectors one vector on the left is

um the difference between the prior and the posterior over action and the other vector is um the expected free energy for each policy and essentially if the two vectors are pointing in the same direction for an organism that's bad news right because um if my posterior is um results in larger expected per energy then that's uh that's bad because i'm trying to minimize it and then wait let me turn the light because it's looking good um dark room solved what's wrong just solve the dark room and yeah so the the idea is basically that the expected free energy you can think of it as a kind of landscape across the domain of potential actions what i want to know is that um after integrating my perceptual evidence and did i get close like did what that vector look like on the left did it match the original vector i had in terms of the expected energy um that's why if you look closely it's the prior minus the posterior because that's the negative rate of change right it's the posit like the actual rate of change would be the posterior minus the prior but we flipped the sign there such that you get um you can interpret the effective charge as like being positive when the precision goes up being negative when precision goes down is that does that answer your question satisfactory yeah i think that's very helpful thanks one example that came to mind was like you're driving and you're having a maps application tell you it's gonna take one hour and then as the estimate gets worse it's like it's bad if you want to get there if your preference is to get there on time now if your preference was to get there later maybe it is more neutral but then as the update is changing and your relative policy it could be like oh it's going to be 18 more minutes because of this crash up but then this speed it up this way and so we're always in a really specific situation conditioning on policy and the information we're getting and so we're not doing the whole traffic city flow this is about like the person getting the info from the screen and then making decisions about oh maybe i should get gas here or not not every possible decision they could be making so very very interesting way to frame this and this dot products and the way i don't know how many people here familiar with the way this kind of vector calculus uh tends to work that essentially you can just think of it as the dot product match yeah calculating the degree of overlap between the vectors and if they're anti uh if they are exactly in the opposite directions then the dot product is very negative they're pointing in exactly the same direction than this maximally positive um but and are these yeah what are the dimensionalities so you're talking about them as vectors but are they scalars there's just one entry in the vector or are there potentially multiple or how do we think about are these one number or is it a list of numbers like a long vector and so in this order formulation of the expected energy it was like one value per policy essentially so every policy has which is like an action sequence and so

every policy has a total kind of sum of the expected free energy associated with that so then it's a vector in the sense that every policy has its own element in the vector but more recently we've actually extended this in the sophisticated sophisticated inference paper because um i mean anybody who's worked with um policy spaces and knows about this problem of combinatorial explosion that once you start considering um courses of action in time then every time every point in time you expand it further uh the more um yeah the more untenable and intractable the problem becomes but in a more recent iteration we actually sub yeah divided further in terms of the actions components in turn yeah instead of having one single policy vector that has to regulate do all the machinery we have every time step has its own action variable and your performing impedance doing inference on that on every time step and this makes the problem prevents you from having to integrate everything right away the end result of that is again an integration but because here it kind of allows for parallel computing let's speak anyway but that's that's a different paper i i can actually also post it i i think we actually read it but you know it all blurs together but we did read sophisticated inference i think at one point oh yeah yeah nice yeah you tweeted lots of papers yes that's why you have them catch up we gotta catch up you're leaving them as fast as we can get to them i mean i made uh more recently uh in a on called sophisticated effective influence where um so you have another thing to catch up but it's essentially combining this effective inference story that we're talking about today and combining it with this article research and which allows you to simulate an agent that responds has an effective response to imagined futures it would be cool to maybe at some point to come back to discuss that one as well cool all right so dean uh with a question and then anyone else who raises their hands casper when you um go from a 2d m1 m2 to a 3d action is that have you thought a little bit about the sort of the less about the integration more about the deflation and and inflation that happens when you go from fewer dimensions to to to more um yeah so that's kind of one of the motivations for going to new types of generative models because this policy variable here can explode pretty quickly once you get to multiple steps in the future and you need somehow to reduce the dimensionality of the state space that you're considering and then what i kind of what kind of one thought that kind of suits me when i build these types of models is that even though um as we say all models are wrong but some are useful right but even though that's the case we know that the organisms that we're trying to model have exactly the same problem that we have namely they're trying to make something that's generally intractable tractable so the simplifications when we're trying to build a model of their model of the world and you can actually justify the

simplifications in the sense that they are trying to deal with they have to do oversimplifications as well to be able to make sense any sense at all with their empowerment i mean i don't know if this exactly answered your question but this was part of it came up in response this year had an interesting point about the dimensionality increasing one thing that kind of happens is that you basically have to stimulate potential futures internally once you start conditioning these days on your actions that means every possible action has a parallel kind of process of inference that's happening um to predict what's going to happen in the future in every dimension you add on top can potentially um make the whole thing interactable so we're always kind of facing that challenge yeah dean and then that's quick follow-up so so sometimes you hear the expression oh that problem is going to require some outside the box thinking and of course that's so ambiguous nobody really knows what that means but what you've done here i think is giving people an opportunity to get out of the spatial envelope get out of the i'm an agent and enveloped by the world around me and you've given them a bit of an eyes of this perspective they can actually get outside and see how as an agent they're perceiving the world how they can then potentially act on the world but as you said they're constantly updating a lot of this stuff isn't front of mind as they're doing it um but i think that's one of the big things about going from 2d to 3d you're not zooming in and zooming out you're you're not just now you're able to look down on something but if you hit the button on the google map and you suddenly see it 3d it doesn't it doesn't stay at one position it actually circles around the thing that you're looking at and i think that's what this provides people is that sense of not just being inside of something but actually being able to step outside of it as well i mean that's something that um yeah actually does happen in this type of model is when you add this um pie let's say in m^3 and you end up expanding the whole thing and it's is like you said like stepping from two to three 3d in a sense because you you're you're expanding the whole number of possibilities basically and in the same way it's kind of hard to wrap your mind around it and i think that's part of what makes these figures hard to read for people especially when you present the top level right away it's like asking somebody to step from one from one dimensional to three-dimensional right in one go you kind of have to go through the the increases in dimensionality step by step so this is also part of what motivates this incremental presentation thanks awesome question thank you dean so blue so we read this sophisticated affective inference like a long time ago and i always think about it and i think that i brought it up in the dodge zero live stream actually but i thought it was the big five paper i think we read them like right at the same time and they they're very but this

um sophisticated active affective inference was really um like prevalent with anxiety like the future time steps the further out you go in the future like the more like you can't predict what's gonna happen and so that's like the underlying basis for anxiety and i thought about and i also thought about um the question about motivation and it made me think of tony robbins has like this um like theory that there's like six like driving factors that like influence people or motivate people through life and like you know depending on personality type or whatever and um they're like uh certainty is one of them um uncertainty is also one of them like people who like variety or um then there's like growth and altruism and i can't remember like all of them but what they all are but it's interesting to think about like the motivation what underlies like what's an underlying motivation for people and how that relates into this type of model as well as like those driving forces through life like how you might start to model those things like um you know someone who's driven by uncertainty versus someone who's driven by certainty that like the the uncertainty driven people might place more of like an epistemic value uh than an actual um then an actual um action reward value so it's just interesting to think about that yeah i mean there's like a paper in the works that got installed unfortunately but um i think where we're trying to do something like that where you're talking about um the way the big five um have kind of emerged as apparently a pretty strong factorization people's behavior people's behavioral tendencies over long periods of time so their personalities and why they tend to factorize like that and to which extent you can capture them with something that looks like this and with at least one more layer i would say but to some extent yes you can capture this tendencies towards exploration a kind of exploration drive versus so risk aversion and risk basically risk seeking behaviors um cool blue that was really interesting and it reminded me about how people prefer let's just say to read different amounts and to read different topics so for some people historical fiction versus fiction science fiction all these different genres and sub-genres it's related to maybe what their regime of attention will latch onto and that's something that's different as you age uh and it's in cultured and it's embedded as well so it's kind of an interesting example that we could go into and then we did a live fact check active11 way back when in 2020 we did do uh affective inference the simulating anticipatory responses paper and that was way more anxiety driven no it wasn't just because it was 20 20 and then also the first one of this year with adam saffron and colin deyoung was the big five cybernetic big five kind of free energy inspired or based as well so um yep it's just it's good you know you you you did say we're kind of going through them fast there's so many to read and just to even get a little bit of a grasp on it it's it to read 25 in a year or one every two

weeks it's kind of a pace that we have to hold up to but i'm sure you read more than one every two weeks so yeah yeah i i guess i did um i did read it somewhere and i was like whoa you um i wasn't sure i didn't remember whether it was like sophisticated inference or sophisticated affected intensive but so what is the effective one it was just the conference proceedings though so it was very short um yeah it's a little bit too minimal i think to really um it was more like a technical note yeah it would be hard to really get it just based on if you're not yeah so familiar with the modeling and the actual like nitty-gritty you know like it's like in that sense confidence proceedings are just a way to communicate to some uh people who are just in the exactly in this technical part domain not best for communication to broader scientific audiences yep well kind of on that point and anyone can raise their hand i move the slide to the additional information where there's the code and um maybe just what would you say about the code what are the inputs or the outputs what would somebody need to run this code or like what could they do with it and it's really cool that you did provide all this information but to kind of bridge that gap that you just mentioned what does this function do or what do these different scripts do um yeah so essentially you first need spm 12 is that contains the core functionality for um yeah that it's based on and then these scripts are like additional or like adaptations of scripts in spm-12 and the only thing you need to do is run spm mdp pbx emo app um so that stands for emotions factorized which is something i worked on at some point and but yeah in the in the rest of the so let me see sorry that's of course not enough to for you to and then you just get the model that we already have uh re-run generate here yeah the one that's on this uh slide before so this yeah this one yeah and the one above it so let me just it's not by now it's so long but i would have to um this is cool but um we're in the process of um what i would recommend actually in the future i mean i don't to be honest i don't like mob at all um the only reason i worked with it was the existing spm and active influence modeling was part of the existing spm package but there's a lot of yeah important work has been done by alex chance and connor heinz and a few others and to transform everything to python and to do computational optimization while we're at it so there's this github page called interactively yep here we are on it pymdp infractively is the github repo yes so um instead of trying to get um trying to get motlop running on your pc i would recommend just working with five and then in the future um we'll have a module in there that can do the same as what we did in this um awesome so that's something i think is much more sustainable for the future we're working on integration with gpus etc to make things that can be simulated and scalable in scalable way can integrate with tensorflow so you can even do um kind of hybrid modeling with what they

call amortized active inference it's like the learning models that are connected to components of these lego lego boxes that i was describing but if you want i can i mean i can run you through this mdp in motlock it's just that i'm not a fan of myself so it would be i think it would be a little bit of a yeah i think yep let's uh definitely have a model stream for the python because i know that will be something a lot of people are interested in because the matlab it has a certain charm to it but i can see how you might want to develop with other approaches even though when christopher and ryan walked us in the model stream through the code just seeing how the matrices multiply there's something nice about how it was in matlab but it's not interfacing with all these modern tools like tensorflow that you mentioned um yeah um anyway but uh let me see so i think we can yeah we can move on but uh i think what i should do basically based on your feedback is write um a little bit of an explanation of what to do with those codes if you want to run it let's say you do want to use mobile if you want to run it what you should do and that's actually useful feedback yeah the reason i added these codes was mostly for yeah reproducibility instead in case other technical folks want to look at what i did but and it's not really out of the box in a way that anybody not familiar with it could easily use it yep my advisor she would always say make it so that the anthropologists from mars will know what to do with that spreadsheet because even for people in the field or even yourself months later it's like wait what that was my research yeah yeah i mean the proper proper practice is to add like a how to like a read me page on the github it's in my mind the kind of transitioning away from what i kind of felt like but i do agree should round up that part of the documentation well yeah yeah um blue or steven uh either of you raise your hand um just just one point about the effective charge piece that you mentioned before can i just check is is that like an in-between um level between one inference layer and the next is it so that is that correct that's entirely correct once you go to figure um i think it's figure six um let me see here seven sorry yep so um essentially now we connect um what's happening in a higher level connected to inferences about our sort of predictions about what's happening on the lower level and that's when the effective charge basically becomes like an ascending message that informs the inferences there is an equation in the manuscript i don't see because it was a separate table oh this part we split figure seven yeah this is just this these are the prior so this like um doesn't describe the posterior beliefs the effective charge basically factors yeah comes into yeah i i understand there's many figures in the paper you didn't include that one but if you to it's in in multiple places essentially what we call effective evidence on page files page 420 but the actual page is 23 you know with pdf okay um just for here we go yeah answer your

question and yeah yeah so just to illustrate what's happening uh or to explain and in this paper use bars to illustrate to indicate posterior beliefs so here the bar is on the left the bar s is like posterior beliefs at time capital t so this is the cross trial time steps a large u yeah large yeah cross trial time essentially a subscript the superscript here is indicating the effective state so u state the posterior effective state on the higher level is a soft max function of u that consists of the prior belief which is the logarithm of the previous posterior from the previous time step multiplied by the transition matrix so that's the where the prior comes from and then this term that you see after that comes from what we call bayesian model reduction and it's what you need to what happens when you want to connect a discrete and continuous state space modules and it drops again this term drops out the derivations that you can do based on this u formalism bayesian model reduction and in this case this sort of free energy assuming that changes are small enough locally are small enough to make this connection and then if you ask what's small enough well that's something you would have to test in in the world and see if this approximation holds and essentially what you see here is the effective charge coming back to haunt you so to speak but now as a message that this passed upwards so the higher level state is consists of two u kind of extremes and so the the extreme positive side so β plus and the extreme negative side β minus and at any point in time the organism is somewhere in between it's never extreme it's not it's never like at the exact extreme it's it's beliefs just like you can never have 100 certainty about anything because down downwards in the hierarchy we make predictions that's where we do beijing mobile averaging and upwards you have to gather these messages so that's what i call effective techniques that's where we use bayesian model reduction yeah so that's essentially a very long answer to your question but yeah no thanks just just one other thing with that is this then gives a way for somatic kind of processes u you know in every day or even in trauma to be integrated in a way because it could be non-cognitive processes could be at play in adjusting this effective charge or keeping a score of that u and then that could then be that could give a way for that to influence or action model selection yes and and this high-level state is not constrained in any way in terms of what you you can add other types of evidence from other action models and other components of your system can add their own effective charge so u that's at some point in the beginning we mentioned this affected workspace theory so like we call this effective charge because it's essentially a pretty domain in general it can be gathered from any kind of action model it can also be used to model the way and when you're listening to music your attentional action model has is kind of being played around with and with congruence

and this congruence and you can kind of um create mismatches and matches and you can create fluency and this this fluency and this will create some kind of effective rollercoaster so to speak so these ups and downs um can be gathered from an action model that would be guiding your attention states um so the question of what to attend to when you're listening to a piece of music could be something that generates effective effective charge and then informs your balance state it can also be purely associative so um it doesn't exclude purely associative types of valence so there can be certain things and that comes back in the contextual evidence if you go a little bit down anymore i think it's here oh yeah oh yeah yeah so you can also gather evidence from in this case we just talked about contextual states in terms of whether the food is on the left or on the right but um you can also have contextual being passed back up and that can generate some kind of top poplopolian learning where you just learn to associate particular contexts with particular positive states or negative states so there's also space to include these kind of um what i think are computationally a little bit less interesting types of valence uh paleons experience because they're just kind of happy and learning in a sense and but it doesn't make them less important for um or experience so these purely associative relationships also come actually we discussed that in the there's in the discussion section see how far we get nice i wanted to ask actually about figure 10 what were you showing in figure 10 or how could we read this and does it apply to your model only or future models other categories of models and everything that's happening here from the orange orange level downwards let's say is is what we implemented but everything that's in the grade is basically something we presupposed so we presupposed for example that this rat already learns how to how the mace works and we presuppose that they have some perceptual capacity or instance some action control over let's say where their body is you know all the things that we needed to presuppose in order to actually do the simulation demonstration or uh yeah are in kind of the great great part of this um box um but the nice thing is that once you so what if you follow the arrows um these gray parts so evolution development and learning are it's also comported in terms of time skills right um and the nice thing is that it's always a circular story so you get um these different nested time skills that are all influencing each other recursively and once you get down to our actual computational experiment you you are at the point in time where and you enter basically yeah on the left this grey arrow that points into the affected box it's kind of yeah you can think of that as the one that initializes the simulation so to speak so this gray arrow is what initialized everything that allowed us to even simulate it and then we explore the dynamics of what happens within that and all the other arrows are basically

illustrating that so the orange arrows provide priors for the precision for the action for the perceptual state so that's all the orange arrows then from the meta minimal according to cognition level again there's this arrow pointing down to perform action jointly the effects end up informing perception and then it kind of passes through in the way that you described also last time in the last session and that's where it kind of connects to the world and then it passes back up again and that's where this kind of perceptual integration happens and it trickles upwards into all these layers we could have simulated learning as well so that's where this orange arrow puts points back up on the right to the posterior phenotype we didn't simulate learning because it wasn't the focus of the paper but we have yeah we have the machinery it's in our lego box to add that um and that actually i think will make the story much even more interesting because then you can think about um active learning and how that influences affective states so how people can enjoy learning just for the sake of learning or how they can learn to associate affective states and contextual states and how that time recursively yeah influences their their system there's one simulation that i've been preparing where we have a setup like this and we actually simulated the yeah the development and learning as and the idea there is that you have a child you kind of simulate the child and it has this um hyper parameters that influence different parts of its lower level model but in the beginning these hyper parameters are very unstable and then there is a kind of parent simulated parent that labels the states like um like basically kind of reflecting um and giving the system labels to work with and in the end what's the idea here is that the labels end up stabilizing the inferences and this kind of to simulate the idea as we know that social interaction is crucial to develop any kind of emotional control emotional self-control and so there's these famous stories right about children growing up with involved something like that and then never really reach a level where they can exert this um type of self self control and reflected that's the communicative capacity that we have anyway so that's another tangent very very interesting one yes very interesting stephen and then anyone else with a question and and in this diagram you mentioned so you mentioned minimal metacognition and then you've got effect and context um so would that minimal metacognition be a sort of phenomenological consciousness the kind of the the the the awareness that you can't necessarily take a perspective on and and it goes up into affect and context and that's where it's consolidated enough to be able to sort of take a perspective on it would that be correct or and that's how i do tend to view it to some extent i mean um the actual representation of these precisions um is often uh assumed or hypothesized to be to occur in in a localized fashion in the brain to some extent in this trade

and that's that's just a hypothesis that can be tested and it's actually an interesting part and that other systems other biological systems can have entirely different kinds of representations for this kind of this precision term so daniel works with ants colonies they might have their own way of encoding this type of reliance on action mode and that maybe has to do with weather conditions or something like that there can be some kind of shared very minimal way of encoding that reliance it doesn't it doesn't necessarily it's not in that sense pretty agnostic on how it's represented and you can just test hypotheses and how it works and also um it's really interesting how we can draw out that link to qualitative concepts or to phenomenological experience but that's not part of this model this is a claim about a modeling architecture that lends itself to certain kinds of calculations that maybe previously would have fallen within the domain of information theory or just control theory or cybernetics or just bayesian inference or just multi-scale systems modeling and i know that those are some of the areas that you've drawn on in your work casper which is why you could kind of see that there was kind of one two three but not four five six for where the active inference model was going because you even discussed how before some of your recent papers the active inference model had less temporal depth and less metacognitive depth so it's just really cool to hear about how it kind of remaps where the questions and the modeling approaches fit in with respect to maybe previous disciplinary approaches so dean and then anyone else with a oh you're meter dean yep um so because you've got that the time flow from left to right and that's pretty much consistent and you've got the parabolic introduced from top down and then back up can you see casper the the idea that around perception around action around minimal metacognition and around effect and context there could be a counterclockwise spin um because i actually saw that in the people were working not even aware of this model but that that's kind of the direction where you've got the flow across the bottom and then the introduction of the parabola from top down and then back up and that's where you could get actually get people looking through both directions that's why you have the effect of valence that's part of where the charge came from or am i sort of confusing the matters because i when i looked at that figure i actually saw how people in that flow state were able what direction the spin was actually going around the middle the middle of your um the middle of your of your figure yeah i mean this figure is is more like yeah it's not it kind of moves away from the very precise um mathematical meaning of the other graphs right that direct exactly progress but it's a little bit of the same ideas here like you said um in that sense it doesn't really matter where they're moving i mean the reason that i put the upwards arrows on the right is that in temporal sense um it's it's representing the

posteriors right so you're moving from the prior to the posterior but then if you would um because it's nested every next level means that you already cycled from basically yesterday's posterior is tomorrow's prior earth like yesterday's for serious two days prior in some way right so um the capacity to integrate to the next iteration can also be itself a problem so you see that a lot with people with traumas that actually integrate an experience too strongly in the way they work and this actually comes to very interesting questions where it's like functional forgetting something like a lot of maybe a lot of these meditative practices actually are able to do it's like help us forget the relative the irrelevant bits or the things that we shouldn't take to the next iteration right the counterclockwise spin is just making it expressing reflection and so sometimes you choose to reflect on sometimes you over reflect on it which is what you're describing but all i'm saying is is that when i saw this it was it was just more confirmation of yeah i think it's i think it's correct that's not just one yeah it's a nice idea and how it's uh i mean the reflection part and kind of becoming aware of the things that you want to so how you want to learn basically um it's something that we we did try to capture in and it gets very interesting you start to be able to simulate processes like meditation and what's happening in the mind when you develop control on these attentional processes on a higher level um anyway but i think stephen wants to say that yep yep just even go for it yeah just following on from that and this this what's interesting is how this brings in this idea of effective representation and how we think of representation normally as something as a thing in the head as opposed to maybe objects in the niche which we act on or create or interact with and now you've got an idea in st of st2 i suppose of this kind of what could be stored in an accessible form where the prior phenotype may not be entirely accessible except through science you know is is this effective representation i just wondered how you see because we've had quite a lot of conversations about this is what what do you see representation being in all of this all the possibility for active inference and representation yeah so um what i like about this approach is that we basically don't have to um pin down exactly and what the effective state means as long as we we're just figuring trying to figure out what it is yeah how it can be purged or an organ and the organism can use very limited cues to infer that and then you have like a whole kind of organic state of the system that that will correlate with that inference to some extent but just as the organism is doing is doing that internally um and usually we think about this in terms of animals but it's interesting to think about more abstract types of representation and three just in the end the statistical concept so i work with models where you add a layer on top of this effective contextual layer so like here just in terms of um to link it back to verbal expressions

and if you ask me what does the representation mean in this case it just has a very specific computational effect on how you can think of it as a generative model generative models so each effective state will be correspondent with certain modes of cognition and action and all the way down to the lowest level the model doesn't have to integrate or like specify all of those details as long as you know how let's say the connections from each um in each level happen the rest becomes like an emergent hole and i think i've tried to be sensitive of the way in the internal representations can be are not necessarily the same as the actual effective state if you look at all the different layers combined and that's when you think about that that then you start to get into really domains like this disconnectedness between your beliefs about your effective state and the way your system is currently behaving and the different kinds of affective disorders you can start thinking about basically hallucinations or like self green forests and anxiety as indian one of those i guess i don't really have a very concrete answer because these representations are are meant to be abstract until you start to model their them in specific context and i think oh yeah you want to yeah now i think that that that makes a lot of sense and also if our represents if what i see as being a representation or models that we people think is in people's heads are actually always something in the niche that we work with so to speak so what we know is the action model for how to act and perceive to recapitulate it just feels like we have that also in the brain but if it's actually to some extent this would be the most concrete model is the emotional imprint that helps then undertake the interaction with the models that we work with but those models don't need to be in the brain in representational form but you might need some affective code isn't it possible to know how to and that ties in a lot with some of the work with micro phenomenology and some of the work with um working with mental space psychology so it's quite interesting um yeah yeah so um thanks for this yeah i mean um what motivated me actually to move in this direction is something that's not really included in the figure that much here that's super important it's the social dimension and then once you think of this effective state like kind of internal representations that are inferred on any number of queues and any number of arbitrary number of sources of data you can also think of the facial expressions of micron specifics as being informative my effectively you start having a very natural way of modeling things like contagion and just in general this kind of empathic responses and where if you're also kind of tracking the group effect and that has places priors on your in your individual experience basically have a very natural way of moving from the connecting kind of the bodily uh interaction with the world in the niche as you described to this abstract social relations that we're tracking all the time so i'm involved in

a project right now that's um forging like primate foraging and it's really interesting to think about the group dynamic versus like seeking out your own epistemic knowledge right so this is just foraging and it's you know agent-based modeling project but i i wonder to what extent like instead of knowing which trees have fruit for example like if i could just follow a group member right like so i wonder you know i mean just what you had said about the social dynamic is this kind of interchangeable like just following the group like i mean we're modeling contagion and so forth so following the group is that like interchangeable for um you know epistemic knowledge really um so there's there's different things you can do i mean one one of the ways you could factor that in here it's like where um something that has been worked out in the active influence communities in terms of beyonce queues basically queues that other specifics give you to kind of indicate what context you are and then your action model specifying for this particular context what is the appropriate action so a very simple thing can be like in uh whether you are in a context where you're following another specific or not it can be something that i think bees have their kind of dance that they do to indicate whether other bees should be following them so there's like a kind of way in which these levels can talk to each other through signaling basically the appropriate what the context is and then from that inference about the context you can have the whole action model kind of rolling out on the lower level and i think the following the behavior that you described does have a pretty direct um analogy to our innate tendency to um synchronize objective states with our specifics even with not non yeah even with other animals that's actually pretty something that we that other animals also seem to be able to do to pick up on nervousness across species right so stress markers of stress we seem to be able to kind of pick up on those kind of cross cross species generalized fashion it's like endless number of we can go down here's here's one more little rabbit hole looking at this figure it reminded me of concurrent programming languages like go lang and the idea of having processes that were defined by interfaces so we've been talking a lot on the as markov blankets or holograms with crisp fields but interfaces are also programming patterns that make nested processes that can be effective or tractable or run on distributed computation without halting or spiraling out so uh in addition to the python potentially there could be something like a concurrent implementation and then it relates to this question about representation like is the representation a static object is it a dynamic process and uh i don't know what the formal answer is but it reminds me of what we did talk about with chris fields with the types as processes and the two spaces so maybe by defining these interfaces with the right dimensionality or bandwidth or structure however the screens

are defined then there will potentially be model ability not philosophical clarity on whether oh is the group is the party conscious or the person or the part of the brain it's like the ant question those are gonna be potentially perennial debates but the model ability of the ant colony evacuation is going to be more akin to the model ability of these other higher level processes because of how abstractly but also excessively defined the interfaces are yeah i think in a similar way the way i these representations i'm kind of playing with just thinking of them entirely as implicit and just just descriptions um of something that implicitly occurs in the system basically like a kind of tool essentially um and if you make it explicit it yeah it ends up being amenable to doesn't mean that there's ever this kind of explicit representation in the system actually and all we're doing is making these things explicit so we can actually compute things i'm pretty happy with being agnostic about whether there's anything and i pretty uh i have no doubt in the fight of representationalism versus um like an activism i think they're they're both pretty i mean they both have points and i'm pretty happy with the models being agnostic on this question because cool models now speak for themselves in the end cool it's almost like you're putting up your code and your model as in evidence and if they want to prosecute the case if they want to have their debate and debate whether your evidence is supporting their notion or this notion or some future notion it's a second level question and the first level is actually what you're just laying out here and how the variables are linked to each other and then there's alternate architectures that are possible yeah every every model is in that sense um just a very elaborate hypothesis and you can just test it and like doing empirical work yep and then every architecture has uh yeah that has a can be compared to each once you have um and to do that you need to go explicit and that's the only part i think where we're doing being rejected to make it uh to get a computational grip on it's a scientific grip you could say actually i i have been working on getting more observational constraints in these types of needs but there's such an amazing richness of things to explore by just even just capturing the phenomenology of our that we already have accessible for in the first person that's often you can already get very far by just making sure that whatever your model is going to be it has to be able to recapitulate our life experience that we have directly accessible it's actually one of the more recent paper that i could share is from generative models to generated passages i don't know you are familiar with this kind of work by varela if i can ask one question on this figure casper so you mentioned that models are hypotheses which might be really interesting to people because they might think about the model as generating hypotheses which it also can do but actually you're talking about how the architecture and the way that the

variables are connected is itself not some authoritative claim or final uh description of the system but it's a hypothesis so it makes me wonder if somebody says well i think that there's you know letter um you know insert your favorite letter here and i want to hook it up to e i want e to be influenced by a new letter r or something like that is that going to um keep this nice mathematical tractability are there certain kinds of wires that if you cross them just the script is going to die is there certain pieces where we know that we can build the legos really well or are there just how do we know which pieces can be tinkered with and what structural changes could even be done or is it like all all by all um well one of the constraints is this basically the way it works is that you can have only local interactions and that's the markov blanket so in the end the way it's kind of kept at least close to biological plausibility is by assuring that a certain thing that you only specify um local interactions between variables and that's what these errors do so one kind of forbidden thing to do would be to connect this policy variable directly to the observations because then you're breaking the market as the policies are directing in this case the relations between hidden states and the hidden states are generating or like based on the hidden states you're generating observations predictions about them um so there are certain kind of forbidden things in these directed basically graphs that being said it's pretty universally i mean as i said whenever there's an arrow it actually implies bi-directional interaction and you're completely free to add a higher level state that modulates your e matrix so the yeah the sky is the limit so to speak i mean in the end you should be able as long as you keep this markov blanket structure intact so you ensure that it's biologically possible what you're doing then you can do yeah you can make any number of states connect to each other if there's like redundancy in your model that should show up because the free energy can be decomposed as accuracy minus sorry complexity minus accuracy so any increase any increase in the number of states in your model will result in increasing uh increase in uncertainty about the parameters and that's the complexity so like occam's razor is built in to the way these models work so yeah in some yes you're free to do almost anything here but if you look at the the model evidence in the end should we punish you with your major model and necessarily complex thanks for this awesome response we'll have a question from dave and then any closing thoughts and then that will be it for 19.1 so dave go for it yeah two points about the point of local interactions only they're kind of at odds with one another first is that actually the definition of locality whatever interacts immediately is by definition local second in one of the computer simulations that was put together at the fristan lab several years ago the dots on the screen where he's trying to get i know it's i assume a simple model of

how market microphone blankets work he found that the distant portions of the simulation were more accurately modeled than the nearby ones and in my mathematical illiteracy i cry out oh so it's all working holographically the intermediate entities are focusing the more distant ones yeah so this is really interesting that you picked up on this because and this is why we kind of um we start with this markov blanket criterion but then when you start um when you kind of go into the way these dynamics work actually it gets more interesting once you're able to predict things that are um outside of your system and the further away the harder it gets usually um but like you described um in this so i think you're talking about the markov blanket the kind of soup the emergence emergency the emergent life paper and in that case for some reason it also has to do with the way you set up the state space system but the way he identified the cro basically he did it just in terms of cross-correlation between the states of these chemical particles and the locations of the state of the particles outside and it was an emergent effect i'm not sure i think it has to do with the particular simulation setup but if you if we kind of abstract it a little bit away from that particular simulation when you go to deep temporal models and what you're trying to do is you're trying to you could call it the hologram i guess in some way you're trying to create an image of what the future looks like outside of the current blanket and that's why they call these the temporal models instance semi-markovian so lots of these interesting cognitive phenomena actually emerge when you start hack your way out of your blanket so to speak and you're trying to get a grip on what's happening outside and that can be spatially it can be temporary so temporary will be in the future especially with the places you can't reach so yeah that's a very interesting question i think um the seminar open aspect it's something that has been actually discussed recently i think it holds an important key to the way our cognition works is that by kind of absorbing information over time and combining with the right priors and the right so nature and nurture have to be like kind of coalescing in a way that gives you a grip on things that are not accessible anyway so that's a very interesting question that i can't directly answer because it's um we just know that for any physical system to like a biological system to work we can only assume local interactions and then the rest of the story is trying to figure out how far you can get it's just local to create those what you call holograms i guess you could call them lego is there a particular literature you're drawing on can you use that term i think chris field actually talked about holograph specifically dave what were you thinking of holograms okay well i got that from the chapter of mark solms book that just came out in the last number of weeks uh where he first is talking about visiting uh carl fristen but he doesn't use the term holographic

the only place i've seen that in this kind of context is a some discussion of um using interferometry to see around corners uh this is something that presumably people can't do but instruments can that there's a holographic or holograph like effect that um allows both either with sound or with light to image objects that can't be seen visibly you have to have a corner as long as there's a corner evidently that induces the kind of self-interference that allows distant uh perception but you know more more relevant probably is the way that con specifics in flocks and so forth uh convey information mediated information sorry i don't have anything more specific awesome awesome points steve thanks for sharing it um it's time and that was an awesome point one discussion really appreciate having you on here casper and everyone else who's joining live so we kind of just close with little you know pause the video or pause time whatever your affordances think about these questions see you and thanks for organizing yep oh for sure and just fill out the feedback form in the calendar invite uh we'll see you next week so keep thinking about the paper and other topics we'll be back for a follow-up discussion on the same paper next week so bye everyone bye